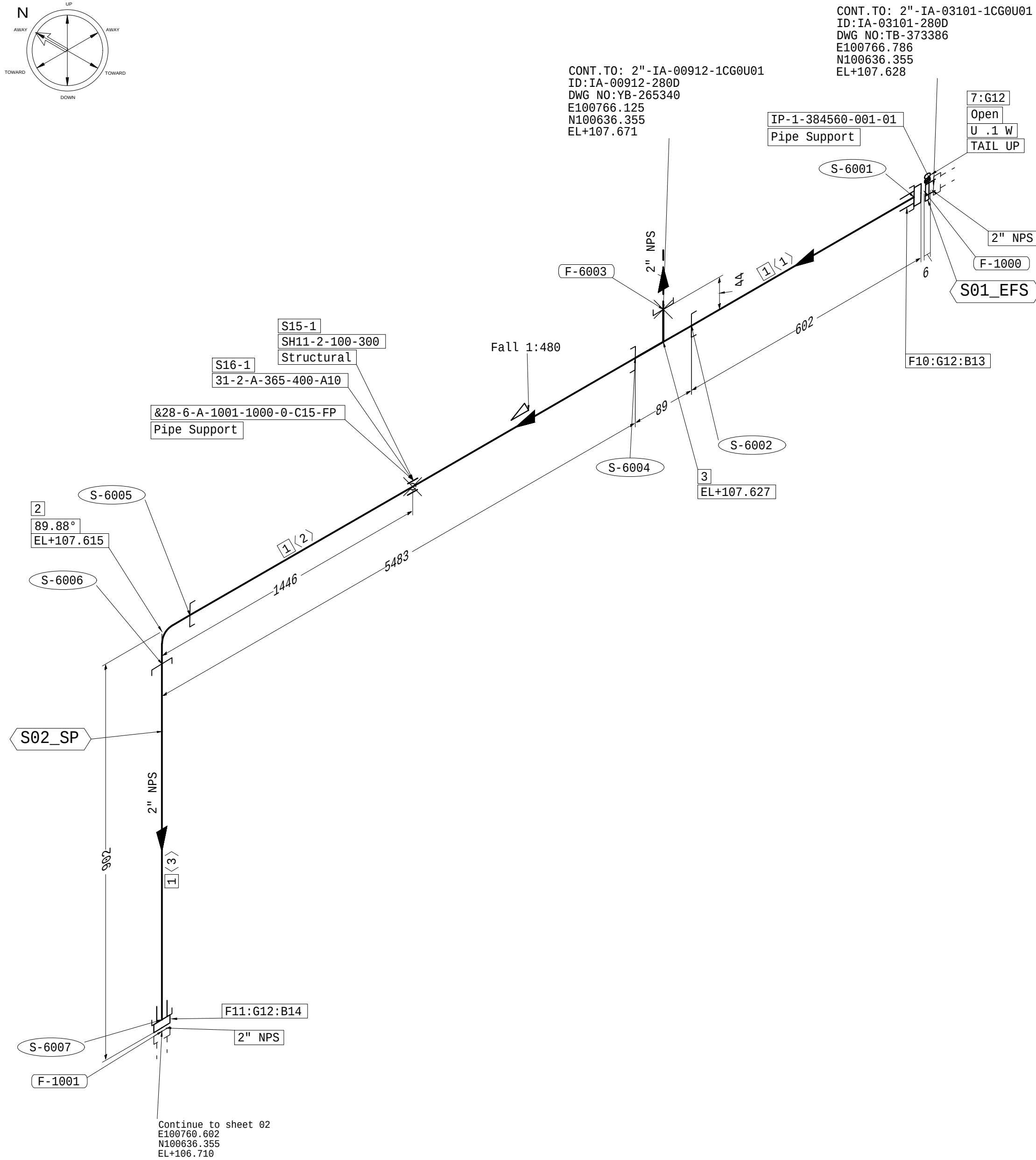
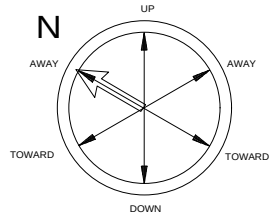


THIS SPOOL MARKUP DRAWING IS GENERATED FROM ENGINEERING PDF & REFERRING APPROVED ENGG ISOMETRICS. ENGINEERING ISOMETRICS TO BE REFERRED FOR REVISION APPROVAL INFORMATION & IN CASE OF ANY DISCREPANCY DATA GIVEN IN ENGINEERING ISOMETRIC SHALL GOVERN



All Pipe attached supports shall be welded before galvanising.

NOTES CONT. :

91. FOR SOUR SERVICE PWHT IS MANDATORY FOR 1. DOUBLE SIDED WELDING, 2. INTERNAL WELD REPAIR 3. INTERNAL WELD BUILD UP FOR DISSIMILAR THICKNESS JOINTS TO OVERCOME MISALIGNMENT.

PWHT	CHEMICAL CLEANING	PIPING MATERIAL CLASS	DESIGN		OPERATING		TEST PRESS. (PSIG)	TEST MEDIUM	RADIOGRAPHY %	INSULATION		TRACING	PAINTING	STRESS SYS NO.	LINE ID
			TEMP. (DEG. F)	PRESS. (PSIG)	TEMP. (DEG.F)	PRESS. (PSIG)			SHOP	FIELD	THK. (MM)	TYPE			
N	NOTE-8	1CG0U01	185	145	135	125	125	SERVICE	NA	NA	0	NI	NA	APCS-26	IA-00901-280D

MATERIALS

PT No.	DESCRIPTION	NPD(IN)	ITEM CODE	QTY
PIPES				
1	PIPE SMLS 2" A106-B GALV STD NPT (M) B36.10M	2	PPS2AEAE02ZZABUAUA	6.898
FITTINGS				
2	ELB 90° FRG LR 2" A105N GALV 3000# NPT (F) B16.11	2	PE72ADAE02ZZAA0ATA	1
3	TEE EQ. FRG 2" A105N GALV 3000# NPT (F) B16.11	2X2	PTE2ADAE02ZZAA0ATA	1
7	SPCT. BLD FF 2" A516 GR70 NORMALISED GALV 150# 125 SPEC	2	PLF2RVAE02ZZAAAADZ	1
FLANGES				
10	FLG SCRD RF JACK SCREW 2" A105N GALV 150# 125 B16.5	2	PLN2ADAE02ZZAAAADH	1
11	FLG SCRD RF 2" A105N GALV 150# 125 B16.5	2	PFH2ADAE02ZZAAAADH	1
GASKET				
12	GSK SPR WND W/CS CENTERING RING + I - SS 2" 4.50 MM THICK SS316L+GRPH FILL 150# B16.20	2	PGUUB5AG02AIAAAZZC	3
FASTSNERS				
13	BOLT, STUD+2 NUTS A193 B7/ A194 2H, CERAMIC FLP COAT B18.2.1/18.2.2 - 100 mm Length	0.625	PH8AB5AGAF00100ZZF	4
14	BOLT, STUD+2 NUTS A193 B7/ A194 2H, CERAMIC FLP COAT B18.2.1/18.2.2 - 90 mm Length	0.625	PH8AB5AGAF00090ZZF	4
SUPPORTS				
15	PIPE SUPPORT	2	SH11-2-100-300	1
16	PIPE SUPPORT	2	31-2-A-365-400-A10	1

PIPE CUT LENGTH SUMMARY

SPL No.	NPD	PC No.	LENGTH	ALLOWANCE	TOTAL LENGTH
S02_SP	2	<3>	0.858	0.000	0.858
S02_SP	2	<2>	5.438	0.000	5.438
S02_SP	2	<1>	0.602	0.000	0.602

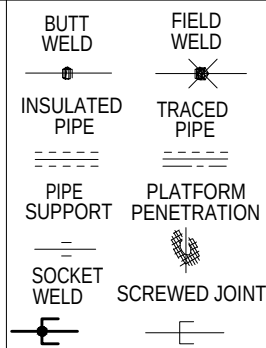
SXX_Spool sequence Number (S02,02,
SXX_SP - Fab. Spool
SXX_PP - Fab. P to P Doubling (c-0M+0M)
SXX_RP - Erec. Random Pipe (0M,12M)
SXX_LP - Erec. Part Loose Fittings
SXX_LF - Erec. Flange (F-Loose Flange,
FTR-Bolt, FTS-Spectro)S
SXX_EV - Erec. valve
SXX_EPG - Erec. Pipe Plug
SXX_SBF - Small Bore Spool Fab.
SXX_SBE - Small Bore Spool Erec.
SXX_SBD - Small Bore Spool Drain
SXX_SBV - Small Bore Spool Vent

X PART NUMBER
<X> PIECE NUMBER
<S> SHOP WELD
<F> FIELD WELD
<SXX_01> SPOOL NUMBER

ISO For

NOTES :

- ALL COORDINATES AND ELEVATIONS ARE IN 'METER' AND ALL DIMENSIONS ARE IN 'MM'.
- PIPE QUANTITY IS IN METERS AND OTHER QUANTITY IN NUMBERS.
- FOR STANDARD SUPPORT, REFER DOC. NO. "PP-866-ST-0001"
- IN CASE OF ANY DISCREPANCY FOR LINE CONDITIONS, LINE DATA GIVEN IN LINE LIST SHALL GOVERN
- FOR SMALL BORE LINES, DIMENSION & ROUTING SHALL BE FIELD ADJUSTED.
- THE SPECIFIED THEORETICAL LOCATION OF EQUIPMENT NOZZLES, SUPPORTING STRUCTURE SHALL BE VERIFIED BEFORE INSTALLING THE PIPING & INTOLERABLE DISCREPANCIES SHALL BE DISCUSSED WITH DESIGN SITE ENGINEER.
- ALL THREADED CONNECTION SHALL BE SEAL WELDED, EXCEPT DRAIN AND VENT PLUG.
- REFER LINE LIST.
- ALL PRESSURE WELDS, OTHER THAN BUTT WELDS, INCLUDING BRANCH CONNECTIONS SUCH AS BRANCH-OLETS, AND BOSS) SHALL BE 100% PTMT EXAMINED. ALL ATTACHMENT WELDS ON PRESSURIZED COMPONENTS THAT ARE NOT HYDROTESTED AFTER MAKING THE ATTACHMENT WELD SHALL BE 100% PTMT EXAMINED.



STANDARD DETAIL REFERENCE DRAWINGS

DETAILS HEAVY WELDING BOSSSES FOR THREADED CONNECTIONS
CONSTRUCTION DETAILS AND DETAILS FOR ISOMETRIC PIPING
INSULATION DETAILS FOR HOT PIPING
INSULATION DETAILS FOR COLD PIPING
INSTALLATION OF JACK SCREWS FOR FLANGED JOINTS

AE-036375
AD-036377
AA-036911
AA-036913
AC-036630

NOTES

- MATERIAL SHOWN WITH A BROKEN LINE AND SPOOLS LISTED IN THE SAUDI ARAMCO FABRICATION LIST ARE NOT TO BE FURNISHED BY THE VENDOR.
- PIPING DETAILS SHALL BE CUT AT ALL POINTS MARKED F.W. (FIELD WELD).
- EACH FABRICATION SECTION SHALL BE CLEARLY MARKED WITH ITS CODE NUMBER.
- UNLESS OTHERWISE NOTED: (A) ALL DIMENSIONS ARE TO FOR FACE OF FLANGES. (B) BOLT HOLES ON FLANGES STRADDLE C. (C) WELDING ELBOWS ARE LONG RADIUS.
- FABRICATION SHALL BE IN ACCORDANCE WITH SAUDI ARAMCO SPECIFICATION 01-SAMSS-10.
- GASKET ALLOWANCE
- PIPING MATERIAL IS

PIPE SIZE	SCHEDULE	SPECIFICATION

9. WELDING STRESS RELIEF - YES ☐ NO ☐ SPOT X-RAY 01-SAMSS-10 ☐
- 100% X-RAY ☐
9. TESTING SHOP HYDROSTATIC TEST - YES ☐ NO ☐ THE SHOP HYDROSTATIC TEST SHALL BE CARRIED OUT ON PIECES HAVING ALL ENDS FLANGED. HYDROSTATIC TEST PRESSURE: _____
10. ALL PIPING MATERIAL IS IN ACCORDANCE WITH SAUDI ARAMCO ENGINEERING SPECIFICATION SAES-L-105 CLASS: _____ ALL EXCEPTIONS ARE NOTED ON THE ISOMETRIC.
11. INSULATION REQUIRED - YES ☐ NO ☐ INSULATION SHOWN ON DRAWING: _____

SPOOLS TO BE FABRICATED BY VENDOR

NO. PIECES	MARK NO.	SIZE	WEIGHT LBS	NO. PIECES	MARK NO.	SIZE	WEIGHT LBS

SPOOLS TO BE FABRICATED BY SAUDI ARAMCO

NO. PIECES	MARK NO.	SIZE	NO. PIECES	MARK NO.	SIZE

REFERENCE DRAWINGS

PIPING PLAN	
P&ID NO.	YA-221234-001
PLOT PLAN KEY PLAN NO.	TA-293108
INDEX FOR PIPING ISO (AG)	PP-865-LI-0003
INDEX FOR PIPING ISO (UG)	PP-865-LI-0004

00	29-09-25	ISSUED FOR CONSTRUCTION	AC/SN	KY/PD	BSP	HJ
REV	DATE	DESCRIPTION	PRPD BY	CHKD BY	APP BY	APP BY
REVISION STATUS						
REVISION VALIDATION						
THIS REVISION NO 00 IS COVERED, FOR ALL APPROVAL AND CERTIFICATION REQUIREMENTS, PER SAEP-33A, UNDER DRAWING COMPLETION CERTIFICATE NO:						
10-14815-0003-000						

DESIGN CERTIFICATION (DESIGN AGENCY'S) ENGG./DATE		REVIEW FOR KEY DRAWINGS (SAUDI ARAMCO) CERTIFIED BY DATE CONST. AGENCY DATE OPERATIONS DATE		OTHER BY DATE	
REV. NO.	DATE	REVISION DESCRIPTION			

JOB ORDER/REV NO. 10-14815-0003
THIS REVISION IS NOT TO BE USED FOR CONSTRUCTION UNLESS CERTIFIED AND DATED

SAUDI ARABIAN OIL COMPANY

ISOM 2" -IA-00901-1CG0U01

ISO

IA SERVICE

JFGP GAS PROCESSING TRAIN-2

JAFURAH

SAUDI ARABIA

PLANT NO.	INDEX	DRAWING NUMBER	SHT. NO.	REV. NO.
865	L	YB-265450	010f06	00-001

SITE REV. NO.	001	RESTRICTED	SMART	CADD-03
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